

MAE 4730

Lecture 23

10/19/2018

Special cases : checking

1) $E_{\text{total}} = \text{constant}$ ($E_{\text{total}} - E_{\text{total}0}$)

2) Checking constraints (DAE)

3) Simplify to simple pendulum

a) $d_2 = 0$ first link is simple pendulum
 $\ddot{\theta}_2 = 0 \Rightarrow \text{constant } \dot{\theta}$

b) $m_2 = \varepsilon, I_2 = \varepsilon \quad \varepsilon \ll 1$

c) $L_1, m_1, I_1 = 0$ 2 simple independent pendulums

4) Look at animation